Large Language Models Explore by Latent Distilling

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Research Questions

- How can we encourage semantic diversity rather than just lexical variation in LLM test-time scaling?
- Can online latent distillation provide effective novelty signals for exploration during generation?
- How can we implement semantic exploration with minimal computational overhead in production settings?

Core Idea

Standard sampling methods for LLMs produce lexically different but semantically redundant responses, limiting test-time scaling effectiveness. ESamp addresses this by training a lightweight "Latent Distiller" online that predicts deep-layer representations from shallow-layer ones. When the distiller makes high prediction errors, it indicates the model is exploring novel semantic regions. ESamp uses this prediction error as a novelty signal to reweight token probabilities, steering generation toward less-explored semantic patterns while maintaining generation quality.

Key Formula

Optimal KL-regularized policy

$$\pi^*(z|s) \propto \pi_{ref}(z|s)\exp\left(\frac{1}{\alpha}r(s,z)\right)$$

π*(z|s) is the optimal sampling policy, πref(z|s) is the base LLM policy, r(s,z) is the novelty reward based on distiller prediction error, α controls exploration strength. This reweights base model probabilities exponentially by novelty to encourage semantic exploration.

ESamp sampling distribution

$$\pi_{new}(z|s) \propto \frac{\pi_{ref}(z|s)^{1+\beta}}{q_{dist}(z|s)^{\beta}}$$

πnew(z|s) is the final sampling distribution, πref(z|s) is the base model distribution, qdist(z|s) is the distiller-predicted distribution, β controls exploration intensity. This geometric interpolation suppresses tokens that the distiller can predict well (indicating redundancy).

Key Figure

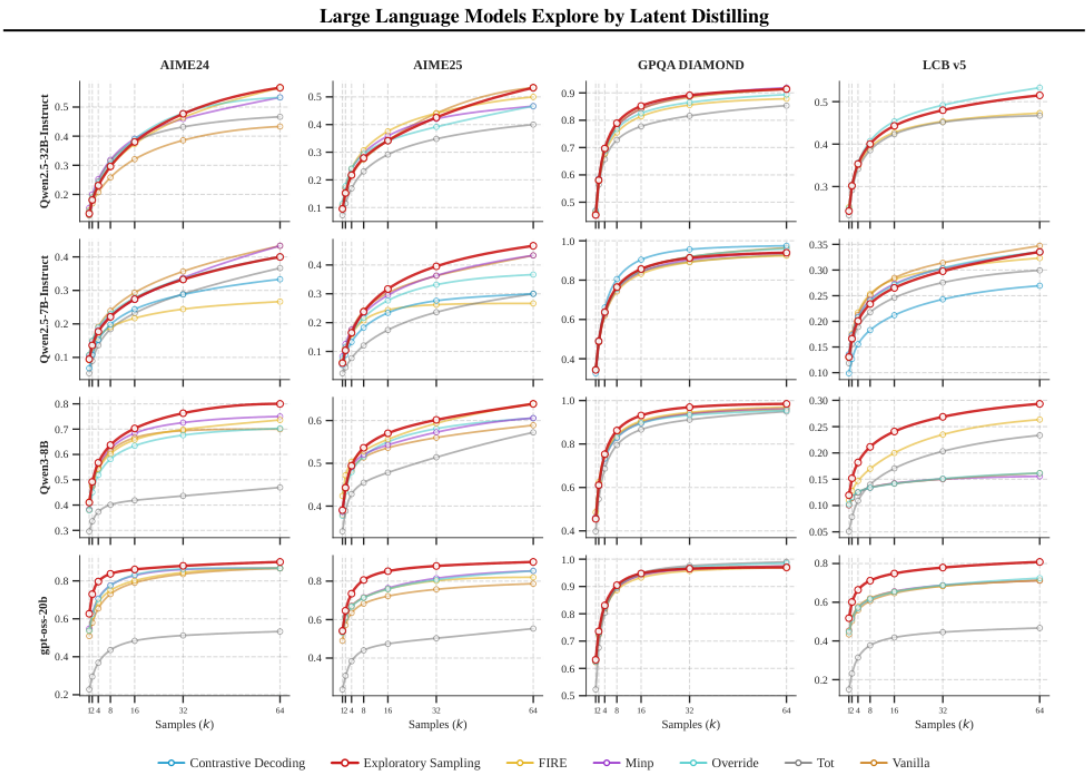

Figure 3. Pass@k performance scaling across different models and benchmarks. ESamp shows superior or comparable performance to all baselines.

Figure 3: Shows Pass@k performance scaling across different models and benchmarks, demonstrating ESamp's superior or comparable performance to all baselines with particular strength in reasoning tasks.

Limitations

- Requires additional memory for the lightweight distiller (though minimal at ~200MB for 8B models)
- Performance depends on the vanishing redundancy assumption which is only approximately satisfied in practice
- May show diminishing benefits when base model already produces highly diverse outputs
- Asynchronous implementation complexity may limit adoption in some serving frameworks